Core Java Practical Slips Solution

Slip 1

1)Write a Java program to find addition of Two Numbers using Command Line Arguments.

```
class Addition {
   public static void main(String args[]) {
      int num1 = Integer.parseInt(args[0]);
      int num2 = Integer.parseInt(args[1]);

      int sum = num1 + num2;

      System.out.println("Addition of two numbers = " + sum);
   }
}
```

2) Write a Java program to define a class Book with data members bookId, bookName, and price. Store details of 5 books using an array of objects and display it.

```
import java.util.Scanner;
class Book {
   int bookId;
   String bookName;
   double price;
   void acceptDetails(Scanner sc) {
      System.out.print("Enter Book ID: ");
      bookId = sc.nextInt();

      sc.nextLine();
```

```
        System.out.print("Enter Book Name: ");
        bookName = sc.nextLine();
        System.out.print("Enter Book Price: ");
        price = sc.nextDouble();
    }
    void displayDetails() {
        System.out.println("Book ID   : " + bookId);
        System.out.println("Book Name : " + bookName);
        System.out.println("Price     : Rs. " + price);
        System.out.println("---------------------------------");
    }
}
public class BookDetails {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        Book books[] = new Book[5];
        System.out.println("Enter details of 5 books:");
        for (int i = 0; i < 5; i++) {
            System.out.println("\nEnter details for Book " + (i + 1));
            books[i] = new Book();
            books[i].acceptDetails(sc);
        }
        System.out.println("\nDetails of all books:");
        System.out.println("---------------------------------");
        for (int i = 0; i < 5; i++) {
            books[i].displayDetails();
        }
        sc.close();
```

```
    }
}
```

OR

2) Write a java program to define a class person(pid, pname, age, gender). Define default and parameterized constructor. Overload the constructor. Accept the 5 person details and display it. (use this keyword).

```
import java.util.Scanner;
class Person {
    int pid;
    String pname;
    int age;
    String gender;
    Person() {
        this.pid = 0;
        this.pname = "Unknown";
        this.age = 0;
        this.gender = "Unknown";
    }
    Person(int pid, String pname, int age, String gender) {
        this.pid = pid;
        this.pname = pname;
        this.age = age;
        this.gender = gender;
    }
    void display() {
        System.out.println("Person ID : " + pid);
        System.out.println("Name     : " + pname);
        System.out.println("Age      : " + age);
        System.out.println("Gender   : " + gender);
```

```
        System.out.println("-----------------------------");
    }
}
public class PersonDetails {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        Person persons[] = new Person[5];
        System.out.println("Enter details of 5 persons:");
        for (int i = 0; i < 5; i++) {
            System.out.println("\nEnter details for Person " + (i + 1));
            System.out.print("Enter Person ID: ");
            int pid = sc.nextInt();
            sc.nextLine();
            System.out.print("Enter Person Name: ");
            String pname = sc.nextLine();
            System.out.print("Enter Age: ");
            int age = sc.nextInt();
            sc.nextLine();
            System.out.print("Enter Gender: ");
            String gender = sc.nextLine();
            persons[i] = new Person(pid, pname, age, gender);
        }
        System.out.println("\nDetails of 5 Persons:");
        System.out.println("============================");
        for (int i = 0; i < 5; i++) {
            persons[i].display();
        }
        sc.close();
```

```
  }
}
```

Slip 2

1) Write a java program to accept n names of cites from user and display them in descending order.

```
import java.util.Scanner;
public class CityNames {
  public static void main(String args[]) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter number of cities: ");
    int n = sc.nextInt();
    sc.nextLine();
    String cities[] = new String[n];
    for (int i = 0; i < n; i++) {
      System.out.print("Enter city " + (i + 1) + ": ");
      cities[i] = sc.nextLine();
    }
    for (int i = 0; i < n - 1; i++) {
      for (int j = i + 1; j < n; j++) {
        if (cities[i].compareToIgnoreCase(cities[j]) < 0) {
          String temp = cities[i];
          cities[i] = cities[j];
          cities[j] = temp;
        }
      }
    }
```

```
        System.out.println("\nCities in Descending Order:");
        for (int i = 0; i < n; i++) {
            System.out.println(cities[i]);
        }
        sc.close();
    }
}
```

2) Write a Java program to derive a class Square from class Rectangle. create one more class Circle. Create an interface with only one method called area(). Implement this interface in all classes. Include appropriate data members and constructors in all classes. Write a java program to accept details of Square, Circle & Rectangle and display the area.

```
import java.util.Scanner;

interface Area {
    void area();
}

class Rectangle implements Area {
    double length;
    double breadth;

    Rectangle(double length, double breadth) {
        this.length = length;
        this.breadth = breadth;
    }

    public void area() {
```

```
        System.out.println("Area of Rectangle = " + (length * breadth));
    }
}

class Square extends Rectangle implements Area {
    Square(double side) {
        super(side, side);
    }

    public void area() {
        System.out.println("Area of Square = " + (length * length));
    }
}

class Circle implements Area {
    double radius;

    Circle(double radius) {
        this.radius = radius;
    }

    public void area() {
        System.out.println("Area of Circle = " + (Math.PI * radius * radius));
    }
}

public class ShapeArea {
    public static void main(String args[]) {
```

```
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter length of Rectangle: ");
    double length = sc.nextDouble();

    System.out.print("Enter breadth of Rectangle: ");
    double breadth = sc.nextDouble();

    System.out.print("Enter side of Square: ");
    double side = sc.nextDouble();

    System.out.print("Enter radius of Circle: ");
    double radius = sc.nextDouble();

    Rectangle rectangle = new Rectangle(length, breadth);
    Square square = new Square(side);
    Circle circle = new Circle(radius);

    System.out.println("\nAreas of Shapes:");
    System.out.println("-------------------------");

    rectangle.area();
    square.area();
    circle.area();

    sc.close();
}
```

}

OR

2) Write a java program to create a Hash table containing Employee name and Salary. Display the details of the hash table. Also search for a specific Employee and display Salary of that Employee.

```
import java.util.Hashtable;
import java.util.Scanner;

public class EmployeeHashTable {
   public static void main(String args[]) {

      Scanner sc = new Scanner(System.in);

      Hashtable<String, Double> employees = new Hashtable<String, Double>();

      System.out.print("Enter number of employees: ");
      int n = sc.nextInt();
      sc.nextLine();

      for (int i = 0; i < n; i++) {
         System.out.print("Enter Employee Name: ");
         String name = sc.nextLine();

         System.out.print("Enter Salary: ");
         double salary = sc.nextDouble();
         sc.nextLine();

         employees.put(name, salary);
      }
```

```
        System.out.println("\nEmployee Details:");
        System.out.println("----------------------------");

        for (String name : employees.keySet()) {
            System.out.println("Employee Name : " + name);
            System.out.println("Salary       : " + employees.get(name));
            System.out.println("----------------------------");
        }

        System.out.print("\nEnter Employee Name to Search: ");
        String searchName = sc.nextLine();

        if (employees.containsKey(searchName)) {
            System.out.println("Employee Name : " + searchName);
            System.out.println("Salary       : " + employees.get(searchName));
        } else {
            System.out.println("Employee not found.");
        }

        sc.close();
    }
}
```

Slip 4

1)Write a java program to check whether a given number is Positive or negative.

```
import java.util.Scanner;
```

```
public class PositiveNegative {
   public static void main(String args[]) {

      Scanner sc = new Scanner(System.in);

      System.out.print("Enter a number: ");
      int num = sc.nextInt();

      if (num > 0) {
         System.out.println("The number is Positive.");
      } else if (num < 0) {
         System.out.println("The number is Negative.");
      } else {
         System.out.println("The number is Zero.");
      }

      sc.close();
   }
}
```

2) Write a java program to define a class CricketPlayer(name, no_of_innings, no_of_times_notout, total_runs, bat_avg). Create an array of "n" player objects. Calculate the batting average for each player using a static method avg (). Handle appropriate exception while calculating average. Define static method "sortPlayer" which sort the array on the basis of average. Display the player details in sorted order.

```
import java.util.Scanner;

class CricketPlayer {
```

```
String name;
int no_of_innings;
int no_of_times_notout;
int total_runs;
double bat_avg;

CricketPlayer(String name, int no_of_innings, int no_of_times_notout,
        int total_runs) {
  this.name = name;
  this.no_of_innings = no_of_innings;
  this.no_of_times_notout = no_of_times_notout;
  this.total_runs = total_runs;
  this.bat_avg = 0;
}

static double avg(CricketPlayer p) {
  int outs = p.no_of_innings - p.no_of_times_notout;

  if (outs <= 0) {
    throw new ArithmeticException("Cannot calculate average");
  }

  return (double) p.total_runs / outs;
}

static void sortPlayer(CricketPlayer players[]) {
  for (int i = 0; i < players.length - 1; i++) {
    for (int j = i + 1; j < players.length; j++) {
```

```
            if (players[i].bat_avg > players[j].bat_avg) {
                CricketPlayer temp = players[i];
                players[i] = players[j];
                players[j] = temp;
            }
          }
        }
    }

    void display() {
        System.out.println("Name           : " + name);
        System.out.println("No. of Innings   : " + no_of_innings);
        System.out.println("Not Out        : " + no_of_times_notout);
        System.out.println("Total Runs      : " + total_runs);
        System.out.println("Batting Average   : " + bat_avg);
        System.out.println("----------------------------------------");
    }
}

public class CricketPlayerDemo {
    public static void main(String args[]) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of players: ");
        int n = sc.nextInt();
        sc.nextLine();
```

```
CricketPlayer players[] = new CricketPlayer[n];

for (int i = 0; i < n; i++) {

  System.out.println("\nEnter details of Player " + (i + 1));

  System.out.print("Enter Player Name: ");
  String name = sc.nextLine();

  System.out.print("Enter Number of Innings: ");
  int innings = sc.nextInt();

  System.out.print("Enter Number of Times Not Out: ");
  int notout = sc.nextInt();

  System.out.print("Enter Total Runs: ");
  int runs = sc.nextInt();
  sc.nextLine();

  players[i] = new CricketPlayer(name, innings, notout, runs);

  try {
    players[i].bat_avg = CricketPlayer.avg(players[i]);
  } catch (ArithmeticException e) {
    players[i].bat_avg = 0;
    System.out.println("Exception: " + e.getMessage());
  }
```

```
        }

        CricketPlayer.sortPlayer(players);

        System.out.println("\nPlayers Sorted According to Batting Average");
        System.out.println("=========================================");

        for (int i = 0; i < n; i++) {
            players[i].display();
        }

        sc.close();
    }
}
```

OR

2) Write a package for String operation which has two classes Con and Comp. Con class concatenates two strings, and comp class compares two strings. Also display proper message on execution.

Step 1- mkdir -p JavaProject/StringOperation

Step 2- cd JavaProject

Step 3- create a package

vim StringOperation/Con.java

```
package stringoperation;
 public class Con {
 public void concatenate(String str1, String str2)
 {
 String result = str1 + str2;
```

```
 System.out.println("Concatenated String: " + result);
 }
 }
```

Step 4- vim StringOperation/Comp.java

```
package stringoperation;
 public class Comp {
 public void compare(String str1, String str2)
{
if (str1.equals(str2))
{
 System.out.println("Both strings are equal.");
 }
 else
{
 System.out.println("Both strings are not equal.");
 }
 }
 }
```

Step 5: Main Program

vim StringOperationDemo.java

```
import java.util.Scanner;
 import stringoperation.Con;
 import stringoperation.Comp;
public class StringOperationDemo {
public static void main(String args[])
{
 Scanner sc = new Scanner(System.in);
 System.out.print("Enter first string: ");
```

```
String str1 = sc.nextLine();
 System.out.print("Enter second string: ");
 String str2 = sc.nextLine();
 Con con = new Con();
Comp comp = new Comp();
System.out.println("\nString Operations:");
System.out.println("-------------------------");
 con.concatenate(str1, str2);
 comp.compare(str1, str2);
 sc.close();
 }
 }
```

Slip 5

1) Write a java program to display the contents of a file in reverse order.

Step 1: create txt file= vim test.txt

Step 2: Main program vim ReverseFile.java

```
import java.io.FileReader;
import java.io.IOException;
import java.util.Scanner;

public class ReverseFile {
   public static void main(String args[]) {

      Scanner sc = new Scanner(System.in);

      System.out.print("Enter file name: ");
      String fileName = sc.nextLine();
```

```
        String content = "";

        try {
            FileReader fr = new FileReader(fileName);

            int ch;

            while ((ch = fr.read()) != -1) {
                content = content + (char) ch;
            }

            fr.close();

            System.out.println("\nFile contents in reverse order:");
            System.out.println("--------------------------------");

            for (int i = content.length() - 1; i >= 0; i--) {
                System.out.print(content.charAt(i));
            }

        } catch (IOException e) {
            System.out.println("Error: File not found or cannot be read.");
        }

        sc.close();
    }
}
```

2) Write a java program to create base class Employee (empcode,empname). Derive the classes anager(designation,club_dues), Scientist (deptname,publications) from Employee class accept the details of n employees and to display the information.

```
import java.util.Scanner;


class Employee {
   int empcode;
   String empname;


   Employee(int empcode, String empname) {
      this.empcode = empcode;
      this.empname = empname;
   }


   void display() {
      System.out.println("Employee Code : " + empcode);
      System.out.println("Employee Name : " + empname);
   }
}


class Manager extends Employee {
   String designation;
   double club_dues;


   Manager(int empcode, String empname, String designation, double club_dues) {
      super(empcode, empname);
      this.designation = designation;
      this.club_dues = club_dues;
```

```
    }

    void display() {
        super.display();
        System.out.println("Designation   : " + designation);
        System.out.println("Club Dues     : " + club_dues);
        System.out.println("--------------------------------");
    }
}

class Scientist extends Employee {
    String deptname;
    String publications;

    Scientist(int empcode, String empname, String deptname, String publications) {
        super(empcode, empname);
        this.deptname = deptname;
        this.publications = publications;
    }

    void display() {
        super.display();
        System.out.println("Department    : " + deptname);
        System.out.println("Publications  : " + publications);
        System.out.println("--------------------------------");
    }
}
public class EmployeeDemo {
```

```
public static void main(String args[]) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter number of employees: ");
    int n = sc.nextInt();
    sc.nextLine();
    Employee employees[] = new Employee[n];
    for (int i = 0; i < n; i++) {
        System.out.println("\nEmployee " + (i + 1));
        System.out.println("1. Manager");
        System.out.println("2. Scientist");
        System.out.print("Enter choice: ");
        int choice = sc.nextInt();
        sc.nextLine()
        System.out.print("Enter Employee Code: ");
        int empcode = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Employee Name: ");
        String empname = sc.nextLine();
        if (choice == 1) {
            System.out.print("Enter Designation: ");
            String designation = sc.nextLine();
            System.out.print("Enter Club Dues: ");
            double club_dues = sc.nextDouble();
            sc.nextLine();
            employees[i] = new Manager(
                empcode, empname, designation, club_dues
            );
        } else if (choice == 2) {
```

```
            System.out.print("Enter Department Name: ");
            String deptname = sc.nextLine();
            System.out.print("Enter Publications: ");
            String publications = sc.nextLine();
            employees[i] = new Scientist(
               empcode, empname, deptname, publications
            );
          } else {
            System.out.println("Invalid choice.");
            i--;
          }
        }

        System.out.println("\nEmployee Information");
        System.out.println("==============================");
        for (int i = 0; i < n; i++) {
          employees[i].display();
        }
        sc.close();
    }
}
```

OR

2) Write a java program to create an abstract class Shape with methods area & volume. Derive two classes Sphere (radius), Cylinder (radius, height) from it. Calculate area and volume of both. (Use Method Overriding).

```
import java.util.Scanner;

abstract class Shape {
   abstract void area();
```

```
    abstract void volume();
}

class Sphere extends Shape {
    double radius;

    Sphere(double radius) {
        this.radius = radius;
    }

    void area() {
        double a = 4 * Math.PI * radius * radius;
        System.out.println("Area of Sphere = " + a);
    }

    void volume() {
        double v = (4.0 / 3.0) * Math.PI * radius * radius * radius;
        System.out.println("Volume of Sphere = " + v);
    }
}

class Cylinder extends Shape {
    double radius;
    double height;

    Cylinder(double radius, double height) {
        this.radius = radius;
        this.height = height;
```

```
    }

    void area() {
        double a = 2 * Math.PI * radius * (radius + height);
        System.out.println("Area of Cylinder = " + a);
    }

    void volume() {
        double v = Math.PI * radius * radius * height;
        System.out.println("Volume of Cylinder = " + v);
    }
}

public class ShapeDemo {
    public static void main(String args[]) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter radius of Sphere: ");
        double sphereRadius = sc.nextDouble();

        System.out.print("Enter radius of Cylinder: ");
        double cylinderRadius = sc.nextDouble();

        System.out.print("Enter height of Cylinder: ");
        double cylinderHeight = sc.nextDouble();

        Sphere sphere = new Sphere(sphereRadius);
```

```
        Cylinder cylinder = new Cylinder(cylinderRadius, cylinderHeight);


        System.out.println("\nSphere Details");
        System.out.println("----------------------");
        sphere.area();
        sphere.volume();


        System.out.println("\nCylinder Details");
        System.out.println("----------------------");
        cylinder.area();
        cylinder.volume();


        sc.close();
    }
}
```

Slip 7

1)Write a package game which will have 2 classes Indoor & Outdoor. Use a function display () to generate the list of players for the specific game. Use default & parameterized constructor.

Step 1: mkdir -p javaproject/game

cd javaproject

Step 2:vim package/Indoor.java

```
package game;
public class Indoor {
   String game;
   String[] players;
   public Indoor() {
      game = "Table Tennis";
```

```
    players = new String[]{"Rahul", "Amit", "Sneha", "Priya"};
  }
  public Indoor(String game, String[] players) {
    this.game = game;
    this.players = players;
  }
  public void display() {
    System.out.println("Indoor Game: " + game);
    System.out.println("Players:");
    for (String player : players) {
      System.out.println(player);
    }
  }
}
```

Step 3: vim package/Outdoor.java

```
package game;
public class Outdoor {
  String game;
  String[] players;
  public Outdoor() {
    game = "Cricket";
    players = new String[]{"Virat", "Rohit", "Rahul", "Hardik"};
  }

  public Outdoor(String game, String[] players) {
    this.game = game;
    this.players = players;
  }
```

```
    public void display() {
        System.out.println("Outdoor Game: " + game);
        System.out.println("Players:");
        for (String player : players) {
            System.out.println(player);
        }
    }
}
```

Step 4: vim mainprogram.java

```
import game.Indoor;
import game.Outdoor;

public class mainprogram {
    public static void main(String[] args) {
        Indoor i1 = new Indoor();
        Outdoor o1 = new Outdoor();

        i1.display();
        System.out.println();
        o1.display();
        System.out.println();
        String[] indoorPlayers = {"Ajay", "Vijay"};
        String[] outdoorPlayers = {"Sachin", "Virat", "MS Dhoni"};
        Indoor i2 = new Indoor("Chess", indoorPlayers);
        Outdoor o2 = new Outdoor("Football", outdoorPlayers);
        i2.display();
        System.out.println();
```

```
    o2.display();
  }
}
```

Step5: javac game/Indoor.java game/outdoor.java mainprogram.java

Step6: java mainprogram

2)Write a java program to define a class MyDate (day, month, year) with methods to accept and display MyDate object. Accept date as dd,mm,yyyy. Throw user defined exception "InvalidDateException" if the date is invalid. Examples of invalid dates: 12 15 2015, 31 6 1990, 29 2 2001

```
import java.util.Scanner;
class InvalidDateException extends Exception {
  public InvalidDateException(String message) {
    super(message);
  }
}
class MyDate {
  int day, month, year;
  void accept() throws InvalidDateException {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter date (dd mm yyyy): ");
    day = sc.nextInt();
    month = sc.nextInt();
    year = sc.nextInt();
    if (month < 1 || month > 12)
      throw new InvalidDateException("Invalid month!");
    int maxDays;
    switch (month) {
```

```
            case 2:
               if (year % 400 == 0 || (year % 4 == 0 && year % 100 != 0))
                  maxDays = 29;
               else
                  maxDays = 28;
               break;
            case 4:
            case 6:
            case 9:
            case 11:
               maxDays = 30;
               break;
            default:
               maxDays = 31;
         }
         if (day < 1 || day > maxDays)
            throw new InvalidDateException("Invalid day!");
      }
      void display() {
         System.out.println("Date: " + day + "/" + month + "/" + year);
      }
   }
   public class DateDemo {
      public static void main(String[] args) {
         MyDate d = new MyDate();
         try {
            d.accept();
            d.display();
```

```
    } catch (InvalidDateException e) {
        System.out.println("Exception: " + e.getMessage());
    }
  }
}
```

OR

2) Write a java program to define a class student having rollno, name and percentage. Define Default and parameterized constructor. Overload the constructor. Accept the 5 student details and display it. (use this keyword).

```
import java.util.Scanner;
class Student{
int rollno;
String name;
float percentage;
Student(){
rollno=0;
name="Unknown";
percentage=0;
}
Student(int rollno,String name,float percentage){
this.rollno=rollno;
this.name=name;
this.percentage=percentage;
}
void display(){
System.out.println(rollno+"\t"+name+"\t"+percentage);
}
}
public class StudentDemo{
```

```
public static void main(String[] args){
Scanner sc=new Scanner(System.in);
Student[] s=new Student[5];
for(int i=0;i<5;i++){
System.out.println("Enter details of student "+(i+1));
System.out.print("Roll No: ");
int rollno=sc.nextInt();
System.out.print("Name: ");
String name=sc.next();
System.out.print("Percentage: ");
float percentage=sc.nextFloat();
s[i]=new Student(rollno,name,percentage);
}
System.out.println("\nRollNo\tName\tPercentage");
for(int i=0;i<5;i++){
s[i].display();
}
}
}
```

Slip 9

1) Write a java program to calculate area of Cylinder and Circle. (use super keyword).

```
import java.util.Scanner;


class Circle {
   double radius;


   Circle(double radius) {
```

```
        this.radius = radius;
    }

    void area() {
        double result = Math.PI * radius * radius;
        System.out.println("Area of Circle = " + result);
    }
}

class Cylinder extends Circle {
    double height;

    Cylinder(double radius, double height) {
        super(radius);
        this.height = height;
    }

    void area() {
        double result = 2 * Math.PI * radius * (radius + height);
        System.out.println("Area of Cylinder = " + result);
    }

    void display() {
        System.out.println("Radius = " + super.radius);
        super.area();
        area();
    }
}
```

```
public class CylinderCircle {
   public static void main(String args[]) {

      Scanner sc = new Scanner(System.in);

      System.out.print("Enter radius of Circle: ");
      double radius = sc.nextDouble();

      System.out.print("Enter height of Cylinder: ");
      double height = sc.nextDouble();

      Circle c = new Circle(radius);
      Cylinder cy = new Cylinder(radius, height);

      System.out.println("\nCircle Details");
      System.out.println("----------------------");
      c.area();

      System.out.println("\nCylinder Details");
      System.out.println("----------------------");
      cy.display();

      sc.close();
   }
}
```

2) Write a java program to accept ‘n’ integers from the user and store them in a collection. Display them in the sorted order. The collection should not accept

duplicate elements. (Use a suitable collection). Search for a element using predefined search method in the Collection framework.

```
import java.util.Scanner;
import java.util.TreeSet;
public class SortedCollection {
   public static void main(String args[]) {
      Scanner sc = new Scanner(System.in);


      TreeSet<Integer> numbers = new TreeSet<Integer>();
      System.out.print("Enter number of integers: ")
      int n = sc.nextInt();
      System.out.println("Enter " + n + " integers:");
      for (int i = 0; i < n; i++) {
         int num = sc.nextInt();
         numbers.add(num);
      }
      System.out.println("\nElements in Sorted Order:");
      System.out.println(numbers);
      System.out.print("\nEnter element to search: ");
      int search = sc.nextInt();
      if (numbers.contains(search)) {
         System.out.println(search + " is present in the collection.");
      } else {
         System.out.println(search + " is not present in the collection.");
      }
      sc.close();
   }
}
```

OR

2) Write a Java program to create a class MyDate with data members dd, mm, and yy. Implement default and parameterized constructors. Display the date in dd-mm-yy format using this keyword.

```
import java.util.Scanner;
class MyDate {
   int dd;
   int mm;
   int yy;
MyDate() {
      this.dd = 1;
      this.mm = 1;
      this.yy = 2000;
   }
   MyDate(int dd, int mm, int yy) {
      this.dd = dd;
      this.mm = mm;
      this.yy = yy;
   }
   void displayDate() {
      System.out.println("Date: " + this.dd + "-" + this.mm + "-" + this.yy);
   }
}
public class MyDateDemo {
   public static void main(String args[]) {
      Scanner sc = new Scanner(System.in);
      System.out.print("Enter day: ");
      int dd = sc.nextInt();
```

```
        System.out.print("Enter month: ");
        int mm = sc.nextInt();
        System.out.print("Enter year: ");
        int yy = sc.nextInt();
        MyDate date = new MyDate(dd, mm, yy);
        System.out.println("\nDate in dd-mm-yy format:");
        date.displayDate();
        sc.close();
    }
}
```

Slip 14

1) Write a java program to accept names of 'n' cities, insert same into array list collection and display the contents of same array list, also remove all these elements.

```
import java.util.Scanner;
class NumberIsZeroException extends Exception{
NumberIsZeroException(String msg){
super(msg);
}
}
public class PrimeDemo{
public static void main(String[] args){
Scanner sc=new Scanner(System.in);
System.out.print("Enter a number: ");
int n=sc.nextInt();
try{
if(n==0)
throw new NumberIsZeroException("Number is 0");
boolean prime=true;
```

```
if(n<2)
prime=false;
else{
for(int i=2;i<=n/2;i++){
if(n%i==0){
prime=false;
break;
}
}
}
if(prime)
System.out.println(n+" is Prime");
else
System.out.println(n+" is Not Prime");
}
catch(NumberIsZeroException e){
System.out.println(e.getMessage());
}
}
}
```

2) Construct a linked List containing names of colours: red, blue, yellow and orange. Then extend your program to do the following: i. Display the contents of the List using an Iterator ii. Display the contents of the List in reverse order using a ListIterator iii. Create another list containing pink and green. Insert the elements of this list between blue and yellow.

```
import java.util.*;
public class ColorList{
```

```
public static void main(String[] args){
LinkedList<String> list=new LinkedList<String>();
list.add("red");
list.add("blue");
list.add("yellow");
list.add("orange");
        System.out.println("List using Iterator:");
        Iterator<String> i=list.iterator();
while(i.hasNext())
        System.out.println(i.next());
       System.out.println("List in reverse order:");
        ListIterator<String> li=list.listIterator(list.size());
while(li.hasPrevious())
           System.out.println(li.previous());
           LinkedList<String> list2=new LinkedList<String>();
list2.add("pink");
list2.add("green");
list.addAll(2,list2);
              System.out.println("List after inserting pink and green:");
Iterator<String> i2=list.iterator();
while(i2.hasNext())
                System.out.println(i2.next());
}
}
```

OR

2) Write a program to read the contents of "abc.txt" file, Display the contents of file in uppercase as output.

Step 1: vim abc.txt

Add text data in this txt file

Step 2: vim FileUpper.java

```
import java.io.*;
public class FileUpper{
public static void main(String[] args){
try{
FileReader fr=new FileReader("abc.txt");
int ch;
while((ch=fr.read())!=-1){
System.out.print(Character.toUpperCase((char)ch));
}
fr.close();
}
catch(IOException e){
System.out.println(e);
}
}
}
```

Slip 15

1) Write a Java program to reverse a number. Accept the number using command line arguments.

```
public class ReverseNumber{
public static void main(String[] args){
int number;
```

```
int reverse=0;
int remainder;
if(args.length==0){
System.out.println("Please enter a number using command line argument");
return;
}
number=Integer.parseInt(args[0]);
System.out.println("Original Number = "+number);
while(number>0){
remainder=number%10;
reverse=reverse*10+remainder;
number=number/10;
}
System.out.println("Reversed Number = "+reverse);
}
}
```

2) Create a class circle (member – radius), cylinder (members – radius, height) which implements this interface. Calculate and display the area and volume.

```
import java.util.Scanner;

interface Shape{
void calculate();
void display();
}

class Circle implements Shape{
double radius;
```

```
double area;

Circle(){
radius=0;
}

Circle(double radius){
this.radius=radius;
}

public void calculate(){
area=3.14*radius*radius;
}

public void display(){
System.out.println("Radius of Circle = "+radius);
System.out.println("Area of Circle = "+area);
}
}

class Cylinder implements Shape{
double radius;
double height;
double area;
double volume;

Cylinder(){
radius=0;
```

```
height=0;
}

Cylinder(double radius,double height){
this.radius=radius;
this.height=height;
}

public void calculate(){
area=2*3.14*radius*(radius+height);
volume=3.14*radius*radius*height;
}

public void display(){
System.out.println("Radius of Cylinder = "+radius);
System.out.println("Height of Cylinder = "+height);
System.out.println("Area of Cylinder = "+area);
System.out.println("Volume of Cylinder = "+volume);
}
}

public class ShapeDemo{
public static void main(String[] args){
Scanner sc=new Scanner(System.in);

System.out.print("Enter radius of circle: ");
double r1=sc.nextDouble();
```

```
Circle c=new Circle(r1);
c.calculate();
c.display();
System.out.print("Enter radius of cylinder: ");
double r2=sc.nextDouble()
System.out.print("Enter height of cylinder: ");
double h=sc.nextDouble();
Cylinder cy=new Cylinder(r2,h);
cy.calculate();
cy.display();
}
}
```

OR

2) Write a java program to create using Runnable interface to display traffic signal.

```
class TrafficSignal implements Runnable{
public void run(){
try{
System.out.println("Traffic Signal");
System.out.println("RED - STOP");
Thread.sleep(2000);
System.out.println("YELLOW - READY");
Thread.sleep(2000);
System.out.println("GREEN - GO");
Thread.sleep(2000);
}
catch(InterruptedException e){
System.out.println(e);
```

```
}
}
}

public class TrafficDemo{
public static void main(String[] args){
TrafficSignal t=new TrafficSignal();
Thread th=new Thread(t);
th.start();
}
}
```

Slip 16

1) Write a Java program to display Fibonacci series.

```
import java.util.Scanner;

public class Fibonacci {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter the number of terms: ");
      int n = sc.nextInt();

      int a = 0, b = 1;

      System.out.println("Fibonacci Series:");

      for (int i = 1; i <= n; i++) {
```

```
        System.out.print(a + " ");

        int c = a + b;
        a = b;
        b = c;
      }

      sc.close();
   }
}
```

2) Write a program to sort HashMap by keys and display the details before sorting and after sorting.

```
import java.util.*
public class SortHashMap {
   public static void main(String[] args) {
      HashMap<Integer, String> map = new HashMap<>();
      map.put(3, "C");
      map.put(1, "A");
      map.put(5, "E");
      map.put(2, "B");
      map.put(4, "D");
      System.out.println("HashMap before sorting:");
      for (Map.Entry<Integer, String> entry : map.entrySet()) {
         System.out.println(entry.getKey() + " : " + entry.getValue());
      }
      TreeMap<Integer, String> sortedMap = new TreeMap<>(map);
```

```
        System.out.println("\nHashMap after sorting by keys:");
        for (Map.Entry<Integer, String> entry : sortedMap.entrySet()) {
            System.out.println(entry.getKey() + " : " + entry.getValue());
        }
    }
}
```

OR

2) Write a Java program to define an interface "Operation" which has Methods, area(), volume(). Define a constant PI having a value of 3.142.

```
interface Operation {
    double PI = 3.142;
    void area();
    void volume();
}
class Circle implements Operation {
    double radius;
    Circle(double radius) {
        this.radius = radius;
    }
    public void area() {
        System.out.println("Area of Circle = " + (PI * radius * radius));
    }
    public void volume() {
        System.out.println("Volume is not applicable for a circle.");
    }
}
```

Slip 17

1)Define an Interface “Integer” with a abstract method check().Write a Java program to check whether a given number is Positive or Negative.

```
interface Integer {
  void check();
}
class NumberCheck implements Integer {
  int num;

  NumberCheck(int num) {
    this.num = num;
  }

  public void check() {
    if (num > 0)
      System.out.println(num + " is Positive");
    else if (num < 0)
      System.out.println(num + " is Negative");
    else
      System.out.println(num + " is Zero");
  }
}

public class PositiveNegative {
  public static void main(String[] args) {
    NumberCheck n = new NumberCheck(-10);
    n.check();
```

```
    }
}
class Cylinder implements Operation {
    double radius, height;
    Cylinder(double radius, double height) {
        this.radius = radius;
        this.height = height;
    }
    public void area() {
        System.out.println("Surface Area of Cylinder = "
                + (2 * PI * radius * (radius + height)));
    }
    public void volume() {
        System.out.println("Volume of Cylinder = "
                + (PI * radius * radius * height));
    }
}
public class OperationDemo {
    public static void main(String[] args) {
        Circle c = new Circle(5);
        c.area();
        c.volume();
       System.out.println();

        Cylinder cy = new Cylinder(3, 7);
        cy.area();
        cy.volume();
    }
```

```
}
```

2) Write a java program to demonstrate Synchronization Example.

```
class Table {
  synchronized void printTable(int n) {
    for (int i = 1; i <= 5; i++) {
      System.out.println(n * i);

      try {
        Thread.sleep(400);
      } catch (InterruptedException e) {
        System.out.println(e);
      }
    }
  }
}

class Thread1 extends Thread {
  Table t;

  Thread1(Table t) {
    this.t = t;
  }

  public void run() {
    t.printTable(5);
  }
}
```

```
class Thread2 extends Thread {
   Table t;

   Thread2(Table t) {
      this.t = t;
   }

   public void run() {
      t.printTable(10);
   }
}

public class SynchronizationDemo {
   public static void main(String[] args) {

      Table obj = new Table();

      Thread1 t1 = new Thread1(obj);
      Thread2 t2 = new Thread2(obj);

      t1.start();
      t2.start();
   }
}
```

OR

2) Write a program to create link list of integer objects. Do the following:

i. Add element at first position

ii. Delete last element

iii. Display the size of link list

```
import java.util.LinkedList;
public class LinkedListDemo{
public static void main(String[] args){
LinkedList<Integer> list=new LinkedList<>();
list.add(10);
list.add(20);
list.add(30);
list.add(40);
System.out.println("Original LinkedList: "+list);
list.addFirst(5);
System.out.println("After adding at first position: "+list);
list.removeLast();
System.out.println("After deleting last element: "+list);
System.out.println("Size of LinkedList: "+list.size());
}
}
```

Slip 18

1)Write a Java program to find the sum of elements of an array. Also display the array elements in ascending order.

```
import java.util.*;
public class ArraySumSort{
public static void main(String[] args){
int[] a={5,2,8,1,4};
```

```
int sum=0;
for(int i=0;i<a.length;i++)
sum+=a[i];
System.out.println("Sum of elements: "+sum);
Arrays.sort(a);
System.out.println("Array in ascending order:");
for(int i=0;i<a.length;i++)
System.out.print(a[i]+" ");
}
}
```

2) Write a Java Program which defines class Product with data member as id, name and price Store the information of 5products and Display the name of product having minimum price(Use array of object).

```
import java.util.*;
class Product{
int id;
String name;
double price;
Product(int id,String name,double price){
this.id=id;
this.name=name;
this.price=price;
}
}
public class ProductDemo{
public static void main(String[] args){
Product[] p=new Product[5];
p[0]=new Product(1,"Laptop",55000);
p[1]=new Product(2,"Mobile",20000);
```

```
p[2]=new Product(3,"Tablet",15000);
p[3]=new Product(4,"Keyboard",1200);
p[4]=new Product(5,"Mouse",500);
Product min=p[0];
for(int i=1;i<p.length;i++){
if(p[i].price<min.price)
min=p[i];
}
System.out.println("Product having minimum price: "+min.name);
}
}
```

OR

2) Write a java program to accept 'N' student names through command line, store them into the appropriate Collection and display them by using Iterator and ListIterator interface.

```
import java.util.*;
public class StudentNames{
public static void main(String[] args){
ArrayList<String> students=new ArrayList<>();
for(String name:args)
students.add(name);
System.out.println("Using Iterator:");
Iterator<String> it=students.iterator();
while(it.hasNext())
System.out.println(it.next());
System.out.println("Using ListIterator:");
ListIterator<String> lit=students.listIterator();
while(lit.hasNext())
System.out.println(lit.next());
```

```
}
}
```

Slip 19

1) Define an Interface Shape with abstract method area(). Write a java program to calculate an area of Circle and Sphere. (use final keyword).

```
interface Shape{
void area();
}
class Circle implements Shape{
final double PI=3.142;
final double radius;
Circle(double radius){
this.radius=radius;
}
public void area(){
System.out.println("Area of Circle: "+(PI*radius*radius));
}
}
class Sphere implements Shape{
final double PI=3.142;
final double radius;
Sphere(double radius){
this.radius=radius;
}
public void area(){
System.out.println("Area of Sphere: "+(4*PI*radius*radius));
}
```

```
}
public class ShapeDemo{
public static void main(String[] args){
Circle c=new Circle(5);
Sphere s=new Sphere(5);
c.area();
s.area();
}
}
```

2) Write a Java program to store student names and their Roll numbers using an appropriate collection and perform following operations: i. Add a new student and its Roll number (No duplicates) ii. Remove a student from the collection iii. Search for a student name and display the roll number.

```
import java.util.*;
public class StudentCollection{
public static void main(String[] args){
HashMap<String,Integer> students=new HashMap<>();
students.put("Rahul",101);
students.put("Priya",102);
students.put("Amit",103);
System.out.println("Students: "+students);
students.putIfAbsent("Sneha",104);
System.out.println("After adding: "+students);
students.remove("Amit");
System.out.println("After removing Amit: "+students);
String name="Priya";
if(students.containsKey(name))
System.out.println("Roll number of "+name+": "+students.get(name));
else
```

```
System.out.println("Student not found");
}
}
```

OR

2) Write a java program to demonstrate Synchronization Example.

```
class Table{
synchronized void printTable(int n){
for(int i=1;i<=5;i++){
System.out.println(n*i);
try{
Thread.sleep(400);
}catch(InterruptedException e){
System.out.println(e);
}
}
}
}
class Thread1 extends Thread{
Table t;
Thread1(Table t){
this.t=t;
}
public void run(){
t.printTable(5);
}
}
class Thread2 extends Thread{
Table t;
```

```
Thread2(Table t){
this.t=t;
}
public void run(){
t.printTable(10);
}
}
public class SynchronizationDemo{
public static void main(String[] args){
Table obj=new Table();
Thread1 t1=new Thread1(obj);
Thread2 t2=new Thread2(obj);
t1.start();
t2.start();
}
}
```

Slip 20

1) Write a Java program to accept a number using the BufferedReader class and generate the multiplication table of the given number.

```
import java.io.*;
public class MultiplicationTable{
public static void main(String[] args)throws IOException{
BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
System.out.print("Enter a number: ");
int n=Integer.parseInt(br.readLine());
for(int i=1;i<=10;i++)
System.out.println(n+" x "+i+" = "+(n*i));
```

```
}
}
```

2) Write a java program to show Multiple Threads Execution.

```
class Thread1 extends Thread{
public void run(){
for(int i=1;i<=5;i++)
System.out.println("Thread 1: "+i);
}
}
class Thread2 extends Thread{
public void run(){
for(int i=1;i<=5;i++)
System.out.println("Thread 2: "+i);
}
}
class Thread3 extends Thread{
public void run(){
for(int i=1;i<=5;i++)
System.out.println("Thread 3: "+i);
}
}
public class MultipleThreads{
public static void main(String[] args){
Thread1 t1=new Thread1();
Thread2 t2=new Thread2();
Thread3 t3=new Thread3();
t1.start();
```

```
t2.start();
t3.start();
}
}
```

OR

2) Write a Java program to store city names and their STD codes using an appropriate collection and perform following operations:

i. Add a new city and its code (No duplicates)

ii. Remove a city from the collection

iii. Search for a city name and display the code.

```
import java.util.*;
public class CityCode{
public static void main(String[] args){
TreeMap<String,Integer> map=new TreeMap<>();
map.put("Pune",20);
map.put("Mumbai",22);
map.put("Delhi",11);
System.out.println("City and STD Code: "+map);
if(!map.containsKey("Nagpur"))
map.put("Nagpur",712);
System.out.println("After adding: "+map);
map.remove("Delhi");
System.out.println("After removing: "+map);
String city="Pune";
if(map.containsKey(city))
System.out.println("STD Code: "+map.get(city));
else
```

```
System.out.println("City not found");
}
}
```

Slip 21

1)Write a java program to calculate area of Cylinder and Circle.(Use super keyword).

```
class Circle{
double r;
Circle(double r){
this.r=r;
}
double area(){
return 3.142*r*r;
}
}
class Cylinder extends Circle{
double h;
Cylinder(double r,double h){
super(r);
this.h=h;
}
double area(){
return 2*3.142*r*(r+h);
}
}
public class AreaDemo{
public static void main(String[] args){
Circle c=new Circle(5);
```

```
Cylinder cy=new Cylinder(5,10);
System.out.println("Area of Circle: "+c.area());
System.out.println("Area of Cylinder: "+cy.area());
}
}
```

2) Write a Java Program which define class Product with data member as id, name and price Store the information of 5products and Display the name of product having minimum price(Use array of object).

```
class Product{
int id;
String name;
double price;
Product(int id,String name,double price){
this.id=id;
this.name=name;
this.price=price;
}
}
public class ProductDemo{
public static void main(String[] args){
Product[] p=new Product[5];
p[0]=new Product(1,"Laptop",55000);
p[1]=new Product(2,"Mobile",20000);
p[2]=new Product(3,"Tablet",15000);
p[3]=new Product(4,"Keyboard",1200);
p[4]=new Product(5,"Mouse",500);
Product min=p[0];
for(int i=1;i<p.length;i++){
if(p[i].price<min.price)
```

```
min=p[i];
}
System.out.println("Product having minimum price: "+min.name);
}
}
```

OR

2) Define an interface “Operation” which has methods area(),volume(). Define a constant PI having a value 3.142. Create a class circle (member – radius), cylinder (members – radius, height) which implements this interface. Calculate and display the area and volume.

```
interface Operation{
double PI=3.142;
void area();
void volume();
}
class Circle implements Operation{
double radius;
Circle(double radius){
this.radius=radius;
}
public void area(){
System.out.println("Area of Circle: "+(PI*radius*radius));
}
public void volume(){
System.out.println("Volume of Circle: "+(4.0/3.0*PI*radius*radius*radius));
}
}
class Cylinder implements Operation{
double radius,height;
```

```
Cylinder(double radius,double height){
this.radius=radius;
this.height=height;
}
public void area(){
System.out.println("Area of Cylinder: "+(2*PI*radius*(radius+height)));
}
public void volume(){
System.out.println("Volume of Cylinder: "+(PI*radius*radius*height));
}
}
public class OperationDemo{
public static void main(String[] args){
Circle c=new Circle(5);
Cylinder cy=new Cylinder(5,10);
c.area();
c.volume();
cy.area();
cy.volume();
}
}
```

Slip 22

1) Write a Java program to accept a number from user and print all prime numbers upto that number (Use Buffered Reader class).

```
import java.io.*;
public class PrimeNumbers{
public static void main(String[] args)throws IOException{
```

```
BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
System.out.print("Enter a number: ");
int n=Integer.parseInt(br.readLine());
System.out.println("Prime numbers:");
for(int i=2;i<=n;i++){
int count=0;
for(int j=1;j<=i;j++){
if(i%j==0)
count++;
}
if(count==2)
System.out.print(i+" ");
}
}
}
```

2) Write a java program to accept 'N' student names through command line, store them into the appropriate Collection and display them by using Iterator and ListIterator interface.

```
import java.util.*;
public class StudentNames{
public static void main(String[] args){
ArrayList<String> list=new ArrayList<>();
for(String name:args)
list.add(name);
System.out.println("Using Iterator:");
Iterator<String> i=list.iterator();
while(i.hasNext())
System.out.println(i.next());
```

```
System.out.println("Using ListIterator:");
ListIterator<String> li=list.listIterator();
while(li.hasNext())
System.out.println(li.next());
}
}
```

OR

2)) Write a Java program to create LinkedList of String objects and perform the following:

i. Add element at the end of the list

ii. Delete first element of the list

iii. Display the contents of list in reverse order

```
import java.util.*;
public class LinkedListDemo{
public static void main(String[] args){
LinkedList<String> list=new LinkedList<>();
list.add("Apple");
list.add("Banana");
list.add("Mango");
list.addLast("Orange");
System.out.println("After adding at end: "+list);
list.removeFirst();
System.out.println("After deleting first element: "+list);
System.out.println("List in reverse order:");
Iterator<String> i=list.descendingIterator();
while(i.hasNext())
System.out.println(i.next());
```

```
}
}
```

Slip 23

1) Write a java program to calculate area of Cylinder and Circle.(Use super keyword)

```
class Circle{
double r;
Circle(double r){
this.r=r;
}
double area(){
return 3.142*r*r;
}
}
class Cylinder extends Circle{
double h;
Cylinder(double r,double h){
super(r);
this.h=h;
}
double area(){
return 2*3.142*r*(r+h);
}
}
public class AreaDemo{
public static void main(String[] args){
Circle c=new Circle(5);
Cylinder cy=new Cylinder(5,10);
```

```
System.out.println("Area of Circle: "+c.area());
System.out.println("Area of Cylinder: "+cy.area());
}
}
```

2) Write a program to sort HashMap by keys and display the details before sorting and after sorting.

```
import java.util.*;
public class SortHashMap{
public static void main(String[] args){
HashMap<Integer,String> map=new HashMap<>();
map.put(3,"C");
map.put(1,"A");
map.put(5,"E");
map.put(2,"B");
map.put(4,"D");
System.out.println("Before sorting: "+map);
TreeMap<Integer,String> sortedMap=new TreeMap<>(map);
System.out.println("After sorting: "+sortedMap);
}
}
```

OR

2) Write a java program to create using Runnable interface to display traffic signal.

```
class TrafficSignal implements Runnable{
public void run(){
try{
while(true){
System.out.println("RED - STOP");
Thread.sleep(2000);
```

```
System.out.println("YELLOW - READY");
Thread.sleep(1000);
System.out.println("GREEN - GO");
Thread.sleep(2000);
}
}catch(InterruptedException e){
System.out.println(e);
}
}
}
public class TrafficSignalDemo{
public static void main(String[] args){
TrafficSignal signal=new TrafficSignal();
Thread t=new Thread(signal);
t.start();
}
}
```

Slip 24

1) Write a java program to accept n names of cites from user and display them in descending order.

```
import java.util.*;
public class CityNames{
public static void main(String[] args){
Scanner sc=new Scanner(System.in);
System.out.print("Enter number of cities: ");
int n=sc.nextInt();
sc.nextLine();
String[] cities=new String[n];
for(int i=0;i<n;i++)
cities[i]=sc.nextLine();
Arrays.sort(cities,Collections.reverseOrder());
System.out.println("Cities in descending order:");
```

```
for(String city:cities)
System.out.println(city);
}
}
```

2) Create an abstract class Shape with methods area & volume. Derive two classes Sphere (radius), Cylinder (radius, height) from it. Calculate area and volume of both. (Use Method Overriding).

```
abstract class Shape{
abstract void area();
abstract void volume();
}
class Sphere extends Shape{
double radius;
Sphere(double radius){
this.radius=radius;
}
void area(){
System.out.println("Area of Sphere: "+(4*3.142*radius*radius));
}
void volume(){
System.out.println("Volume of Sphere: "+(4.0/3.0*3.142*radius*radius*radius));
}
}
class Cylinder extends Shape{
double radius,height;
Cylinder(double radius,double height){
this.radius=radius;
this.height=height;
}
```

```
void area(){
System.out.println("Area of Cylinder: "+(2*3.142*radius*(radius+height)));
}
void volume(){
System.out.println("Volume of Cylinder: "+(3.142*radius*radius*height));
}
}
public class ShapeDemo{
public static void main(String[] args){
Shape s=new Sphere(5);
Shape c=new Cylinder(5,10);
s.area();
s.volume();
c.area();
c.volume();
}
}
```

OR

2) Write a java program to accept 'N' student names through command line, store them into the appropriate Collection and display them by using Iterator and ListIterator interface.

```
import java.util.*;
public class StudentNames{
public static void main(String[] args){
ArrayList<String> list=new ArrayList<>();
for(String name:args)
list.add(name);
System.out.println("Using Iterator:");
```

```
Iterator<String> i=list.iterator();
while(i.hasNext())
System.out.println(i.next());
System.out.println("Using ListIterator:");
ListIterator<String> li=list.listIterator();
while(li.hasNext())
System.out.println(li.next());
}
}
```

Slip 25

1) Write a Java program to reverse a number. Accept the number using command line arguments.

```
public class ReverseNumber{
public static void main(String[] args){
int n=Integer.parseInt(args[0]);
int rev=0;
while(n!=0){
rev=rev*10+n%10;
n=n/10;
}
System.out.println("Reverse: "+rev);
}
}
```

2) Write a package game which will have 2 classes Indoor & Outdoor. Use a function display() to generate the list of players for the specific game. Use default & parameterized constructor.

Step 1: mkdir -p javaproject/game

cd javaproject

Step 2:vim package/Indoor.java

```
package game;
public class Indoor {
  String game;
  String[] players;
  public Indoor() {
    game = "Table Tennis";
    players = new String[]{"Rahul", "Amit", "Sneha", "Priya"};
  }
  public Indoor(String game, String[] players) {
    this.game = game;
    this.players = players;
  }
  public void display() {
    System.out.println("Indoor Game: " + game);
    System.out.println("Players:");
    for (String player : players) {
      System.out.println(player);
    }
  }
}
```

Step 3: vim package/Outdoor.java

```
package game;
public class Outdoor {
  String game;
  String[] players;
```

```
  public Outdoor() {
    game = "Cricket";
    players = new String[]{"Virat", "Rohit", "Rahul", "Hardik"};
  }

  public Outdoor(String game, String[] players) {
    this.game = game;
    this.players = players;
  }
  public void display() {
    System.out.println("Outdoor Game: " + game);
    System.out.println("Players:");
    for (String player : players) {
      System.out.println(player);
    }
  }
}
```

Step 4: vim mainprogram.java

```
import game.Indoor;
import game.Outdoor;

public class mainprogram {
  public static void main(String[] args) {
    Indoor i1 = new Indoor();
    Outdoor o1 = new Outdoor();

    i1.display();
```

```
        System.out.println();
        o1.display();
        System.out.println();
        String[] indoorPlayers = {"Ajay", "Vijay"};
        String[] outdoorPlayers = {"Sachin", "Virat", "MS Dhoni"};
        Indoor i2 = new Indoor("Chess", indoorPlayers);
        Outdoor o2 = new Outdoor("Football", outdoorPlayers);
        i2.display();
        System.out.println();
        o2.display();
    }
}
```

Step5: javac game/Indoor.java game/outdoor.java mainprogram.java

Step6: java mainprogram

OR

2) Write a Java program to create 2 files F1 and F2.Copy the contents of file F1 by changing the case into file F2.F2 = F1 Also copy the contents of F1 and F2 in F3.F3 = F1+F2 Display the contents of F3.

Step 1: vim F1 .txt

Hello Java

Welcome to Programming

Step 2: vim F2.txt

hELLO jAVA

wELCOME TO pROGRAMMING

Step 3: vim FileDemo.java

```
import java.io.*;
```

```
class FileDemo {
   public static void main(String[] args) {
      try {
          FileWriter fw = new FileWriter("F1.txt");
        fw.write("Hello Java\nWelcome to Programming");
        fw.close();
        FileReader fr = new FileReader("F1.txt");
        FileWriter fw2 = new FileWriter("F2.txt");

        int ch;
        while ((ch = fr.read()) != -1) {
          if (Character.isUpperCase((char) ch))
             fw2.write(Character.toLowerCase((char) ch));
          else if (Character.isLowerCase((char) ch))
             fw2.write(Character.toUpperCase((char) ch));
          else
             fw2.write(ch);
        }
        fr.close();
        fw2.close();
        FileWriter fw3 = new FileWriter("F3.txt");
        fr = new FileReader("F1.txt");
        while ((ch = fr.read()) != -1) {
          fw3.write(ch);
        }
        fr.close();
        fr = new FileReader("F2.txt");
        while ((ch = fr.read()) != -1) {
```

```
            fw3.write(ch);
        }
        fr.close();

        fw3.close();
        FileReader fr3 = new FileReader("F3.txt");
        System.out.println("Contents of F3:");
        while ((ch = fr3.read()) != -1) {
            System.out.print((char) ch);
        }
        fr3.close();
      } catch (IOException e) {
        System.out.println("Error: " + e.getMessage());
      }
    }
}
```